Forces: Connected Particles

Questions 1–21

Question 1. By modelling each object as a particle and ignoring air resistance, draw a diagram showing all of the forces acting on:

a) A car of mass M kg towing a caravan of mass m kg on a rough surface.
b) A train engine of mass M kg pulling two trucks of mass m kg on a rough surface.
c) A helicopter of mass M kg with a person of mass m kg hanging directly below.

Question 2. A car of mass 2000 kg is towing a caravan of mass 500 kg along a straight flat road. The driving force produced by the engine is 600 N. By modelling both the car and caravan as having no resistance to motion, find

a) the acceleration of the car.
b) the tension in the tow bar.

Question 3. A car of mass 2200 kg is towing a trailer of mass 400 kg along a straight flat road. The engine of the car is creating a driving force of 1100 N. The resistance force acting on the car is 72 N and the resistance force acting on the trailer is 16 N. Find

a) the acceleration of the car and trailer.
b) the tension in the tow bar.

Question 4. Two particles, P and Q which are connected by a light, inextensible string have mass 6 kg and 1.5 kg respectively. Particle P is pulled along by a horizontal force of 60 N. Particles P and Q have resistances of 12 N and 3 N acting against them respectively.

a) Find the acceleration of the particles.
b) Find the tension in the string.
c) Explain how the modelling assumptions that the string is light and inextensible have been used.

Question 5. Two particles, P and Q which are connected by a light, inextensible rod have mass 10 kg and 2 kg respectively. Particle P is pushed along a rough horizontal surface by a horizontal force of 30 N. Particles P and Q both experience resistances which are proportional to their masses. Given that they are accelerating at 0.5 ms^{-2}, find the thrust in the rod.

Question 6. Two particles, P and Q which are connected by a light, inextensible string have mass 10 kg and m kg respectively. Particle P is pulled along a smooth horizontal surface by a horizontal force of 120 N which creates an acceleration of 3 ms^{-2}. Find

a) the value of m
b) the tension in the string.

Question 7. Two particles, P and Q which are connected by a light, inextensible string have mass $5m$ kg and $2m$ kg respectively. Particle P is pulled along by a horizontal force of 37.8 N. Particles P and Q have resistances of 6 N and 2.4 N acting against them respectively. Find

a) the acceleration in terms of m
b) the size of the tension in the string.
c) Explain how the modelling assumptions that the string is light and inextensible have been used.

Question 8. A train engine of mass 10000 kg is towing 2 identical trucks of mass 3500 kg along a flat straight track which are experiencing the same resistive forces. The driving force produced by the engine is 25 000 N. They are moving with constant acceleration 1.2 ms^{-2}. The engine has a resistance force of 2400 N acting against it. Find

a) the total resistance acting against each of the trucks.
b) the tension in each of the couplings.

Question 9. A car is towing a caravan. The driving force created by the car is 300 N. A constant resistance force of 180 N acts against the caravan. They move at constant velocity. Find the total resistance acting on the car.

Question 10. A car of mass 1800 kg is towing a caravan of mass 800 kg along a straight flat road. They start at rest and then reach 5 ms^{-1} after having travelled 20 metres. The resistance force acting on the car is 90 N and the resistance force acting on the caravan is 40 N. Find

a) the acceleration of the car and caravan.
b) the driving force by the engine of the car.
c) the tension in the tow bar.

Question 11. Two particles, P and Q which are connected by a light, inextensible string have mass 20 kg and 6 kg respectively. Particle P is pulled by a force of 145.6 N. Particles P and Q have resistances of 60 N and 18 N acting against them respectively. The system starts from rest. Find

a) the acceleration of P.
b) the distance P moves over the first 5 s.
c) the tension in the string between P and Q.

Question 12. A train engine of mass 7000 kg is towing a carriage of mass 2000 kg along a straight flat track. The driving force produced by the engine is 12 600 N. The resistance forces on the train engine and carriage are $700k$ N and $200k$ N respectively.

a) Show that $a = \frac{7}{5} - \frac{k}{10}$
b) Find the tension in the coupling between the train engine and the carriage.

Given they start at rest and reach 3 ms^{-1} in 5 s,

c) find the resistance on the carriage.
d) find the resistance on the train engine.

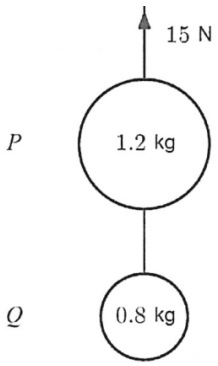
Question 13. A van of mass 2500 kg is towing a trailer of mass 1000 kg along a straight flat road, with a tow bar. The driver reduces the speed of the van from 81 km h^{-1} to 63 km h^{-1} by using the van's brakes, creating a total force of 2590 N against the motion. The resistance force acting on the van is 150 N and the resistance force acting on the trailer is 60 N throughout the motion.

a) find the deceleration of the van.
b) find the time taken to slow to 63 km h^{-1}.
c) find the thrust in the tow bar.
d) State a modelling assumption you have made.

Whilst travelling at 63 km h^{-1} the tow bar breaks. The trailer slows to a rest without a collision. The total resistance to motion to the trailer is now 400 N.

e) find how far the trailer travels before coming to a stop.
f) Explain what the driver of the van would have noticed with the motion of the van.

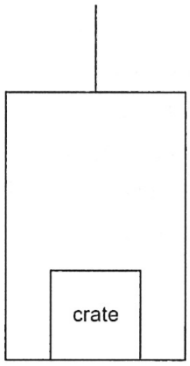
Question 14. The following questions use $g = 9.8$ **ms**$^{-2}$ An object P of mass 1.2 kg is suspended by a light inextensible string. A second object Q of mass 0.8 kg is suspended by a second light inextensible string onto object P, as shown in the diagram. The first string has a force of 15 N applied, and the system moves upwards.



Find:

a) the acceleration of the system.
b) the tension in the lower string.
c) the value the 15 N needs to be increased by to achieve constant velocity.

Question 15. For each of the following questions containing a crate in a lift, the following diagram may be used



A crate of mass m is in a lift which is travelling upwards. Find the magnitude of the force exerted on the crate by the lift when:

a) $m = 10$ kg and an acceleration 0.5 ms^{-2}
b) $m = 100$ g and an acceleration 0.7 ms^{-2}
c) $m = 2$ kg and a deceleration of 1.2 ms^{-2}
d) $m = 450$ g and a deceleration of 0.8 ms^{-2}

Question 16. A lift of mass 800 kg is attached to a light, inextensible cable. It contains a crate of mass 100 kg. Find the tension in the cable when the lift is:

a) Accelerating downwards at 0.5 ms^{-2}
b) Accelerating upwards at 0.5 ms^{-2}
c) Decelerating downwards at 0.4 ms^{-2}
d) Decelerating upwards at 0.4 ms^{-2}

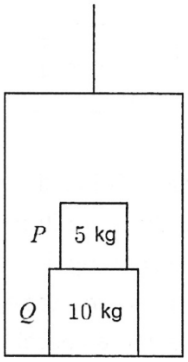
Find the acceleration upwards of the lift when the tension in the cable is:

e) 2700 N
f) 8100 N
g) 9450 N

Question 17. The following questions use $g = 10$ **ms**$^{-2}$ A lift of mass 600 kg is attached to a light, inextensible cable. It contains a crate of mass 50 kg. The lift starts from rest and moves upwards, reaching the required floor which is 10 metres above in 15 seconds. Find

a) the acceleration of the lift.
b) the tension in the cable.
c) the magnitude of the force exerted on the crate by the floor of the lift.

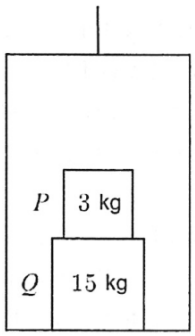
Question 18. A lift of mass 100 kg accelerates upwards at 0.4 ms^{-2}, being pulled by a light inextensible vertical cable. The lift contains two boxes, P and Q of masses 5 kg and 10 kg respectively. Box P sits directly on top of box Q as shown in the diagram below:



Find

a) the tension in the cable.
b) the force exerted on Q by P
c) the force exerted by Q on the lift.

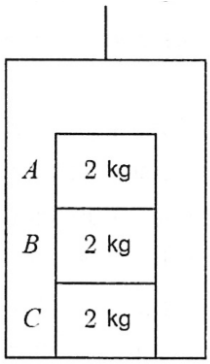
Question 19. A lift of mass 120 kg decelerates upwards at 0.25 ms^{-2}, being pulled by a light inextensible vertical cable. The lift contains two crates, P and Q of masses 3 kg and 15 kg respectively. Crate P sits directly on top of crate Q



Find

a) the tension in the cable.
b) the force exerted on Q by P.
c) the force exerted by Q on the lift.

Question 20. A lift of mass 120 kg accelerates upwards at 0.1 ms^{-2} pulled by a light inextensible vertical cable, carrying three identical crates, A, B and C, each with a mass of 2 kg.



Find

a) the tension in the cable.
b) the force exerted on the lift by C.

Question 21. Prerequisite knowledge for Q21: Variable Acceleration Two particles P and Q which are connected by a light, inextensible string have mass 8 kg and 5 kg respectively. Particle P is being pulled along a smooth horizontal surface with velocity v ms^{-1} at time t seconds given by

$$v = -\frac{1}{5}t^2 + 6t, \quad 0 \leq t \leq 30$$

Find, at 10 seconds into the motion:

a) the acceleration of the particles.
b) the tension in the string.
c) the force pulling P.